**Problem.** Assume $O$ is the midpoint for both segments $[AB]$ and $[CD]$. Prove that $AC = BD$.

Since $O$ is the midpoint for both segments $[AB]$ and $[CD]$, we have $OA = OB$ and $OC = OD$

NO!

Now, $AC \boxed{=} OA + OC = OB + OD \boxed{=} BD$,

hence the result.


$AC \neq AO + OC$